

Esteban Sitek

Senior Software Engineer | TOSCA DI, ETL Testing, Data Quality

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PROFESSIONAL SUMMARY

Senior QA Engineer with **12+** years across healthcare, e-commerce, and fintech, specialized in ETL testing, data validation, and data warehouse quality assurance. Expertise in **Tosca Data Integrity**, complex **SQL** query development, and source-to-target reconciliation across relational and cloud databases. Proven ability to design automated test frameworks, identify data anomalies, and collaborate with cross-functional teams in Agile environments to ensure end-to-end data integrity.

SKILLS

Languages: SQL, Python, TypeScript, JavaScript, HTML, CSS

Testing & QA: Tosca Data Integrity, Tricentis Tosca, ETL Testing, Data Validation, Test Automation, Selenium, Jest, Pytest, Cypress

Backend: Node.js, Express, FastAPI, Django, Spring Boot

Frontend: React, Next.js, Vue.js, Angular, Tailwind CSS

Database: PostgreSQL, MySQL, MongoDB, Elasticsearch, Redis, Snowflake, BigQuery, DynamoDB

Data & ETL: Apache Airflow, Apache Spark, dbt, Kafka, Talend, Informatica, Data Warehouse, Data Lake

Cloud & DevOps: AWS, Google Cloud Platform, Azure, Docker, Kubernetes, Terraform, GitHub Actions, AWS Lambda, RDS, S3, CloudWatch

Others: RESTful API, GraphQL, WebSocket, Git, GitHub, Jira, Confluence, Agile, Scrum, Data Reconciliation, Root Cause Analysis, Defect Analysis

EXPERIENCE

Publicis Sapient
Senior QA Engineer

Jun 2024–Present

- Led the design and development of enterprise data validation frameworks using **Tosca Data Integrity** and **SQL**, automating source-to-target reconciliation across multiple cloud data warehouses and reducing manual validation effort by **40%**.
- Drove comprehensive ETL testing strategy for AI-powered data pipelines, establishing automated test cases in **Tosca DI** that validated business rule transformations, data aggregations, and exception handling across **PostgreSQL**, **BigQuery**, and **Snowflake** environments.
- Championed cross-functional collaboration with platform engineers and data architects to define metadata-based testing protocols, catching data anomalies and integrity violations

before production deployment and eliminating customer-facing discrepancies.

- Developed complex **SQL** queries to perform backend data verification, aggregation validation, and root cause analysis on discrepancies between source systems and target data warehouses, identifying systemic issues affecting 50+ downstream reports.
- Implemented reusable **Tosca DI** test assets and automation libraries that standardized reconciliation patterns across three enterprise client engagements, reducing test case development time by 35% and improving team consistency.
- Conducted end-to-end validation of data migrations involving millions of records, coordinating with developers to trace anomalies back to transformation logic and ensuring data completeness and consistency across source and target systems.
- Established defect classification framework using **Jira** to track and prioritize data quality issues by severity and business impact, enabling the team to resolve critical blockers before data loads to production systems.
- Mentored two junior QA engineers on **Tosca DI** best practices, automated test design patterns, and **SQL** performance optimization techniques for large-scale data validation across distributed architectures.
- Validated data loads into reporting databases and data lakes, automating duplicate detection, missing data identification, and integrity checks using **Tosca Data Integrity** and custom **Python** validation scripts.
- Collaborated with stakeholders to define acceptance criteria for data warehouse migrations, translating business requirements into executable test scenarios that validated reconciliation logic and exception handling pathways.

CareSimple

Senior Full Stack Engineer

Oct 2022–May 2024

- Spearheaded data quality initiatives for healthcare reporting pipelines, designing **ETL Testing** frameworks that validated patient population health metrics, remote monitoring data, and clinical outcomes aggregations using **SQL** and **PostgreSQL** backend validation.
- Built automated test harnesses in **Pytest** and **Jest** to validate end-to-end data flows from IoT patient monitoring devices through cloud ingestion and into analytics dashboards, ensuring data integrity across **AWS Lambda**, **RDS**, and **S3** layers.
- Owned the implementation of reconciliation logic between source patient records and downstream reporting tables, using advanced **SQL** queries to identify missing encounters, duplicate admissions, and aggregation mismatches that affected care metrics.
- Designed and maintained test infrastructure using **Node.js** and **Express** microservices that simulated various data ingestion scenarios, catching transformation bugs before they reached production and causing reporting inaccuracies for hospital partners.
- Collaborated with data engineering team to validate **Apache Airflow** ETL workflows that processed millions of daily clinical records, ensuring correctness of business rule implementations and data completeness across bronze, silver, and gold data layers.

- Developed comprehensive **SQL**-based validation suite for patient data warehouse, performing aggregation testing, exception analysis, and root cause investigation on discrepancies in bed capacity, readmission rates, and treatment timelines.
- Implemented **GraphQL** API contract testing to ensure healthcare data API endpoints returned accurate patient cohorts and aggregated metrics, reducing downstream analytics errors by 28% and improving reporting reliability.

Shopee

Full Stack Engineer

Jul 2018–Sep 2022

- Contributed to the data platform infrastructure supporting 295+ million users, developing **SQL** data validation scripts and test automation frameworks that verified transaction logs, inventory reconciliation, and financial settlement data across regional data warehouses.
- Built distributed backend services in **Node.js** and **Express** that powered order and payment processing pipelines, incorporating comprehensive **SQL** query validation to audit data correctness, duplicate order detection, and transaction reconciliation across **MySQL** and **MongoDB** stores.
- Developed test automation using **Cypress** and **Jest** for e-commerce transaction flows, validating end-to-end data flow from customer purchase through order fulfillment and revenue recognition, identifying edge cases that caused revenue miscalculations.
- Collaborated with data analysts to design and execute **ETL Testing** for the data lake that aggregated seller inventory, customer behavior, and marketplace metrics across Southeast Asian markets, ensuring data consistency for regional reporting.
- Implemented monitoring and alerting using **CloudWatch** and custom **Python** scripts to detect anomalies in transaction data, including duplicate entries and missing reconciliation records, enabling rapid root cause analysis when data quality issues surfaced.
- Optimized **SQL** queries on **Elasticsearch** and **MongoDB** for real-time analytics dashboards, ensuring data validation and completeness checks ran efficiently even during peak transaction volumes of 50k+ orders per second.
- Worked with the fintech team to validate cross-border payment and settlement data flows, implementing reconciliation logic that ensured merchant payouts matched transaction records across currency conversions and fee calculations.

Altitude Games

Full Stack Engineer

Nov 2014–Jul 2018

- Built multiplayer game backend systems in **Node.js** and **Express**, implementing comprehensive **SQL** validation and data integrity checks on player profiles, in-game transactions, and leaderboard rankings stored in **PostgreSQL** and **MongoDB**.
- Designed test automation frameworks using **Selenium** and **Jest** to validate game economy data flows, ensuring player currency transactions, item acquisitions, and progression records were accurate and consistent across backend services.
- Developed analytics data pipeline that aggregated player behavior, session duration, and monetization metrics, implementing custom **SQL** queries and **Python** validation scripts to identify data anomalies and ensure reporting accuracy.

- Owned backend data validation for blockchain-based gaming initiative, verifying asset transactions and NFT metadata consistency between game servers and on-chain records, implementing ETL-style validation workflows.
- Collaborated with gameplay designers to implement automated testing of economy-critical calculations—player rewards, booster mechanics, and progression thresholds—using **SQL** and backend service validation to catch bugs before live deployment.

EDUCATION

Ateneo de Manila University
Bachelor of Science in Computer Science

Aug 2010–Jul 2014